E2 sheet 1 - session and film (one column per session)

Fill at the machine. Write only what you see. Leave blank rather than guess. Session S0 = machine check day; D1, D2 = lasing days.

Item	S0	D1	D2
date			
operator			
lab in charge permission YN			
machine make model			
controller			
tube rating W			
film id lot			
film thickness 1 mm			
film thickness 2 mm			
film thickness 3 mm			
film thickness 4 mm			
film thickness 5 mm			
room C start			
room C end			
RH pct			
chiller C start			
chiller C end			
S ref mm s			
S ref source			
cut power setting			
cut speed mm s			
meter model			
meter max range ohm			
microscope and scale			
notes			

E2 sheet 2 - machine-setting log (copy the values exactly as shown in LaserCAD)

Photograph every layer dialog. 'Min P / Max P' = the two power fields of the layer. Units = what LaserCAD shows next to the field. Scan dir = X or Y.

Coupon	Layer	Planned	Mode	Min P	Max P	Units	Speed	Interval	Passes	Scan dir	Order	Focus	Photo
E2-A-R	RED	Scan 12 @ 50											
E2-A-R	YELLOW	Scan 12 @ 100											
E2-A-R	GREEN	Scan 12 @ 150											
E2-A-R	CYAN	Scan 12 @ 200											
E2-A-R	BLUE	Scan 12 @ 300											
E2-A-R	MAGENTA	Scan 12 @ 400											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
CUTSEARCH	BLUE	Cut ref + 3 per step @ 20											
E2-A-D1-C1	RED	Scan 10 @ S_ref											
E2-A-D1-C1	YELLOW	Scan 12 @ 0.67 x S_ref											
E2-A-D1-C1	GREEN	Scan 12 @ S_ref											
E2-A-D1-C1	CYAN	Scan 12 @ 1.5 x S_ref											
E2-A-D1-C1	BLUE	Scan 14 @ S_ref											
E2-A-D1-C1	MAGENTA	Scan 12 @ S_ref											
E2-A-D1-C1	BLACK (OUTLINE_CUT)	Cut or OFF cut setting @ cut speed											
E2-A-D1-C2	RED	Scan 10 @ S_ref											

E2-A-D1-C2	YELLOW	Scan 12 @ 0.67 x S_ref											
E2-A-D1-C2	GREEN	Scan 12 @ S_ref											
E2-A-D1-C2	CYAN	Scan 12 @ 1.5 x S_ref											
E2-A-D1-C2	BLUE	Scan 14 @ S_ref											
E2-A-D1-C2	MAGENTA	Scan 12 @ S_ref											
E2-A-D1-C2	BLACK (OUTLINE_CUT)	Cut or OFF cut setting @ cut speed											
E2-A-D2-C1	RED	Scan 10 @ S_ref											
E2-A-D2-C1	YELLOW	Scan 12 @ 0.67 x S_ref											

E2 sheet 2 - machine-setting log (copy the values exactly as shown in LaserCAD)

Photograph every layer dialog. 'Min P / Max P' = the two power fields of the layer. Units = what LaserCAD shows next to the field. Scan dir = X or Y.

Coupon	Layer	Planned	Mode	Min P	Max P	Units	Speed	Interval	Passes	Scan dir	Order	Focus	Photo
E2-A-D2-C1	GREEN	Scan 12 @ S_ref											
E2-A-D2-C1	CYAN	Scan 12 @ 1.5 x S_ref											
E2-A-D2-C1	BLUE	Scan 14 @ S_ref											
E2-A-D2-C1	MAGENTA	Scan 12 @ S_ref											
E2-A-D2-C1	BLACK (OUTLINE_CUT)	Cut or OFF cut setting @ cut speed											
E2-A-D2-C2	RED	Scan 10 @ S_ref											
E2-A-D2-C2	YELLOW	Scan 12 @ 0.67 x S_ref											
E2-A-D2-C2	GREEN	Scan 12 @ S_ref											
E2-A-D2-C2	CYAN	Scan 12 @ 1.5 x S_ref											
E2-A-D2-C2	BLUE	Scan 14 @ S_ref											
E2-A-D2-C2	MAGENTA	Scan 12 @ S_ref											
E2-A-D2-C2	BLACK (OUTLINE_CUT)	Cut or OFF cut setting @ cut speed											
E2-B-D1-C1	GREEN (ENGRAVE)	Scan reference @ S_ref											
E2-B-D1-C1	BLUE (CUT)	Cut cut setting @ cut speed											
E2-B-D1-C2	GREEN (ENGRAVE)	Scan reference @ S_ref											
E2-B-D1-C2	BLUE (CUT)	Cut cut setting @ cut speed											
E2-B-D2-C1	GREEN (ENGRAVE)	Scan reference @ S_ref											
E2-B-D2-C1	BLUE (CUT)	Cut cut setting @ cut speed											

E2 sheet 3 - resistance (2-wire DMM)

Silver paint on pads, dry 10 min. 3 readings per strip; lift and re-place the probes each time. Write OL if over range, and the range.

Sample	L mm	W mm	R1 ohm	R2 ohm	R3 ohm	Range	Contact
E2-A-D1-C1-RED-L08	8	2.0					
E2-A-D1-C1-RED-L16	16	2.0					
E2-A-D1-C1-RED-L24	24	2.0					
E2-A-D1-C1-YELLOW-L08	8	2.0					
E2-A-D1-C1-YELLOW-L16	16	2.0					
E2-A-D1-C1-YELLOW-L24	24	2.0					
E2-A-D1-C1-GREEN-L08	8	2.0					
E2-A-D1-C1-GREEN-L16	16	2.0					
E2-A-D1-C1-GREEN-L24	24	2.0					
E2-A-D1-C1-CYAN-L08	8	2.0					
E2-A-D1-C1-CYAN-L16	16	2.0					
E2-A-D1-C1-CYAN-L24	24	2.0					
E2-A-D1-C1-BLUE-L08	8	2.0					
E2-A-D1-C1-BLUE-L16	16	2.0					
E2-A-D1-C1-BLUE-L24	24	2.0					
E2-A-D1-C1-MAGENTA-L08	8	2.0					
E2-A-D1-C1-MAGENTA-L16	16	2.0					
E2-A-D1-C1-MAGENTA-L24	24	2.0					
E2-A-D1-C2-RED-L08	8	2.0					
E2-A-D1-C2-RED-L16	16	2.0					
E2-A-D1-C2-RED-L24	24	2.0					
E2-A-D1-C2-YELLOW-L08	8	2.0					
E2-A-D1-C2-YELLOW-L16	16	2.0					
E2-A-D1-C2-YELLOW-L24	24	2.0					
E2-A-D1-C2-GREEN-L08	8	2.0					
E2-A-D1-C2-GREEN-L16	16	2.0					
E2-A-D1-C2-GREEN-L24	24	2.0					

E2-A-D1-C2-CYAN-L08	8	2.0					
E2-A-D1-C2-CYAN-L16	16	2.0					
E2-A-D1-C2-CYAN-L24	24	2.0					
E2-A-D1-C2-BLUE-L08	8	2.0					
E2-A-D1-C2-BLUE-L16	16	2.0					
E2-A-D1-C2-BLUE-L24	24	2.0					
E2-A-D1-C2-MAGENTA-L08	8	2.0					
E2-A-D1-C2-MAGENTA-L16	16	2.0					
E2-A-D1-C2-MAGENTA-L24	24	2.0					

E2 sheet 3 - resistance (2-wire DMM)

Silver paint on pads, dry 10 min. 3 readings per strip; lift and re-place the probes each time. Write OL if over range, and the range.

Sample	L mm	W mm	R1 ohm	R2 ohm	R3 ohm	Range	Contact
E2-A-D2-C1-RED-L08	8	2.0					
E2-A-D2-C1-RED-L16	16	2.0					
E2-A-D2-C1-RED-L24	24	2.0					
E2-A-D2-C1-YELLOW-L08	8	2.0					
E2-A-D2-C1-YELLOW-L16	16	2.0					
E2-A-D2-C1-YELLOW-L24	24	2.0					
E2-A-D2-C1-GREEN-L08	8	2.0					
E2-A-D2-C1-GREEN-L16	16	2.0					
E2-A-D2-C1-GREEN-L24	24	2.0					
E2-A-D2-C1-CYAN-L08	8	2.0					
E2-A-D2-C1-CYAN-L16	16	2.0					
E2-A-D2-C1-CYAN-L24	24	2.0					
E2-A-D2-C1-BLUE-L08	8	2.0					
E2-A-D2-C1-BLUE-L16	16	2.0					
E2-A-D2-C1-BLUE-L24	24	2.0					
E2-A-D2-C1-MAGENTA-L08	8	2.0					
E2-A-D2-C1-MAGENTA-L16	16	2.0					
E2-A-D2-C1-MAGENTA-L24	24	2.0					
E2-A-D2-C2-RED-L08	8	2.0					
E2-A-D2-C2-RED-L16	16	2.0					
E2-A-D2-C2-RED-L24	24	2.0					
E2-A-D2-C2-YELLOW-L08	8	2.0					
E2-A-D2-C2-YELLOW-L16	16	2.0					
E2-A-D2-C2-YELLOW-L24	24	2.0					
E2-A-D2-C2-GREEN-L08	8	2.0					
E2-A-D2-C2-GREEN-L16	16	2.0					
E2-A-D2-C2-GREEN-L24	24	2.0					

E2-A-D2-C2-CYAN-L08	8	2.0					
E2-A-D2-C2-CYAN-L16	16	2.0					
E2-A-D2-C2-CYAN-L24	24	2.0					
E2-A-D2-C2-BLUE-L08	8	2.0					
E2-A-D2-C2-BLUE-L16	16	2.0					
E2-A-D2-C2-BLUE-L24	24	2.0					
E2-A-D2-C2-MAGENTA-L08	8	2.0					
E2-A-D2-C2-MAGENTA-L16	16	2.0					
E2-A-D2-C2-MAGENTA-L24	24	2.0					

E2 sheet 3 - resistance (2-wire DMM)

Silver paint on pads, dry 10 min. 3 readings per strip; lift and re-place the probes each time. Write OL if over range, and the range.

Sample	L mm	W mm	R1 ohm	R2 ohm	R3 ohm	Range	Contact
E2-B-D1-C1-W030	16	0.3					
E2-B-D1-C1-W050	16	0.5					
E2-B-D1-C1-W080	16	0.8					
E2-B-D1-C1-W100	16	1.0					
E2-B-D1-C1-W150	16	1.5					
E2-B-D1-C2-W030	16	0.3					
E2-B-D1-C2-W050	16	0.5					
E2-B-D1-C2-W080	16	0.8					
E2-B-D1-C2-W100	16	1.0					
E2-B-D1-C2-W150	16	1.5					
E2-B-D2-C1-W030	16	0.3					
E2-B-D2-C1-W050	16	0.5					
E2-B-D2-C1-W080	16	0.8					
E2-B-D2-C1-W100	16	1.0					
E2-B-D2-C1-W150	16	1.5					

E2 sheet 4 - line width, gap and hole size (microscope)

Lines and gaps: 3 positions (left, centre, right of the 16 mm length). Holes: 2 perpendicular diameters. Note the scale used.

Sample	Feature	Drawn mm	Pos 1 mm	Pos 2 mm	Pos 3 mm	Magnif.	Photo
E2-A-D1-C1-RED-L16	line	2.0					
E2-A-D1-C1-YELLOW-L16	line	2.0					
E2-A-D1-C1-GREEN-L16	line	2.0					
E2-A-D1-C1-CYAN-L16	line	2.0					
E2-A-D1-C1-BLUE-L16	line	2.0					
E2-A-D1-C1-MAGENTA-L16	line	2.0					
E2-A-D1-C2-RED-L16	line	2.0					
E2-A-D1-C2-YELLOW-L16	line	2.0					
E2-A-D1-C2-GREEN-L16	line	2.0					
E2-A-D1-C2-CYAN-L16	line	2.0					
E2-A-D1-C2-BLUE-L16	line	2.0					
E2-A-D1-C2-MAGENTA-L16	line	2.0					
E2-A-D2-C1-RED-L16	line	2.0					
E2-A-D2-C1-YELLOW-L16	line	2.0					
E2-A-D2-C1-GREEN-L16	line	2.0					
E2-A-D2-C1-CYAN-L16	line	2.0					
E2-A-D2-C1-BLUE-L16	line	2.0					
E2-A-D2-C1-MAGENTA-L16	line	2.0					
E2-A-D2-C2-RED-L16	line	2.0					
E2-A-D2-C2-YELLOW-L16	line	2.0					
E2-A-D2-C2-GREEN-L16	line	2.0					
E2-A-D2-C2-CYAN-L16	line	2.0					
E2-A-D2-C2-BLUE-L16	line	2.0					
E2-A-D2-C2-MAGENTA-L16	line	2.0					
E2-B-D1-C1-W030	line	0.3					
E2-B-D1-C1-W050	line	0.5					
E2-B-D1-C1-W080	line	0.8					

E2-B-D1-C1-W100	line	1.0					
E2-B-D1-C1-W150	line	1.5					
E2-B-D1-C2-W030	line	0.3					
E2-B-D1-C2-W050	line	0.5					
E2-B-D1-C2-W080	line	0.8					
E2-B-D1-C2-W100	line	1.0					
E2-B-D1-C2-W150	line	1.5					
E2-B-D2-C1-W030	line	0.3					
E2-B-D2-C1-W050	line	0.5					

E2 sheet 4 - line width, gap and hole size (microscope)

Lines and gaps: 3 positions (left, centre, right of the 16 mm length). Holes: 2 perpendicular diameters. Note the scale used.

Sample	Feature	Drawn mm	Pos 1 mm	Pos 2 mm	Pos 3 mm	Magnif.	Photo
E2-B-D2-C1-W080	line	0.8					
E2-B-D2-C1-W100	line	1.0					
E2-B-D2-C1-W150	line	1.5					
E2-B-D1-C1-G020	gap	0.2					
E2-B-D1-C1-G030	gap	0.3					
E2-B-D1-C1-G050	gap	0.5					
E2-B-D1-C1-G080	gap	0.8					
E2-B-D1-C1-G100	gap	1.0					
E2-B-D1-C1-G150	gap	1.5					
E2-B-D1-C2-G020	gap	0.2					
E2-B-D1-C2-G030	gap	0.3					
E2-B-D1-C2-G050	gap	0.5					
E2-B-D1-C2-G080	gap	0.8					
E2-B-D1-C2-G100	gap	1.0					
E2-B-D1-C2-G150	gap	1.5					
E2-B-D2-C1-G020	gap	0.2					
E2-B-D2-C1-G030	gap	0.3					
E2-B-D2-C1-G050	gap	0.5					
E2-B-D2-C1-G080	gap	0.8					
E2-B-D2-C1-G100	gap	1.0					
E2-B-D2-C1-G150	gap	1.5					
E2-B-D1-C1-H1	hole	2.0					
E2-B-D1-C1-H2	hole	2.0					
E2-B-D1-C1-H3	hole	2.0					
E2-B-D1-C2-H1	hole	2.0					
E2-B-D1-C2-H2	hole	2.0					
E2-B-D1-C2-H3	hole	2.0					

E2-B-D2-C1-H1	hole	2.0					
E2-B-D2-C1-H2	hole	2.0					
E2-B-D2-C1-H3	hole	2.0					

E2 sheet 5 - isolation between two lines (highest resistance range)

Probe pad A (lower left) and pad B (upper right). 2 readings. Then look at the gap under the microscope: any black bridge? Pass needs > 3 Mohm (10 x the reader's 300 kohm 'dry' threshold).

Sample	Gap mm	Reading 1	Reading 2	Range ohm	Bridge seen Y/N	Notes
E2-B-D1-C1-G020	0.2					
E2-B-D1-C1-G030	0.3					
E2-B-D1-C1-G050	0.5					
E2-B-D1-C1-G080	0.8					
E2-B-D1-C1-G100	1.0					
E2-B-D1-C1-G150	1.5					
E2-B-D1-C2-G020	0.2					
E2-B-D1-C2-G030	0.3					
E2-B-D1-C2-G050	0.5					
E2-B-D1-C2-G080	0.8					
E2-B-D1-C2-G100	1.0					
E2-B-D1-C2-G150	1.5					
E2-B-D2-C1-G020	0.2					
E2-B-D2-C1-G030	0.3					
E2-B-D2-C1-G050	0.5					
E2-B-D2-C1-G080	0.8					
E2-B-D2-C1-G100	1.0					
E2-B-D2-C1-G150	1.5					

E2 sheet 6 - visual grade (0 none, 1 brown, 2 patchy black, 3 continuous black, 4 flaking, 5 burn-through)

Grade right after lasing, with a macro photo and a ruler in frame. Tape test only on Part A L08 strips, LAST (destructive).

Sample	Grade	Continuous Y/N	Burn-through Y/N	Tape pass/fail	Photo
E2-A-D1-C1-RED					
E2-A-D1-C1-YELLOW					
E2-A-D1-C1-GREEN					
E2-A-D1-C1-CYAN					
E2-A-D1-C1-BLUE					
E2-A-D1-C1-MAGENTA					
E2-A-D1-C2-RED					
E2-A-D1-C2-YELLOW					
E2-A-D1-C2-GREEN					
E2-A-D1-C2-CYAN					
E2-A-D1-C2-BLUE					
E2-A-D1-C2-MAGENTA					
E2-A-D2-C1-RED					
E2-A-D2-C1-YELLOW					
E2-A-D2-C1-GREEN					
E2-A-D2-C1-CYAN					
E2-A-D2-C1-BLUE					
E2-A-D2-C1-MAGENTA					
E2-A-D2-C2-RED					
E2-A-D2-C2-YELLOW					
E2-A-D2-C2-GREEN					
E2-A-D2-C2-CYAN					
E2-A-D2-C2-BLUE					
E2-A-D2-C2-MAGENTA					
E2-B-D1-C1-W030				n/a	
E2-B-D1-C1-W050				n/a	
E2-B-D1-C1-W080				n/a	

E2-B-D1-C1-W100				n/a	
E2-B-D1-C1-W150				n/a	
E2-B-D1-C1-G020				n/a	
E2-B-D1-C1-G030				n/a	
E2-B-D1-C1-G050				n/a	
E2-B-D1-C1-G080				n/a	
E2-B-D1-C1-G100				n/a	
E2-B-D1-C1-G150				n/a	
E2-B-D1-C2-W030				n/a	
E2-B-D1-C2-W050				n/a	
E2-B-D1-C2-W080				n/a	
E2-B-D1-C2-W100				n/a	
E2-B-D1-C2-W150				n/a	
E2-B-D1-C2-G020				n/a	
E2-B-D1-C2-G030				n/a	
E2-B-D1-C2-G050				n/a	
E2-B-D1-C2-G080				n/a	
E2-B-D1-C2-G100				n/a	
E2-B-D1-C2-G150				n/a	
E2-B-D2-C1-W030				n/a	
E2-B-D2-C1-W050				n/a	
E2-B-D2-C1-W080				n/a	
E2-B-D2-C1-W100				n/a	
E2-B-D2-C1-W150				n/a	
E2-B-D2-C1-G020				n/a	
E2-B-D2-C1-G030				n/a	
E2-B-D2-C1-G050				n/a	
E2-B-D2-C1-G080				n/a	
E2-B-D2-C1-G100				n/a	
E2-B-D2-C1-G150				n/a	

E2 sheet 7 - controls (every measuring session)

Same meter and leads as the samples.

Session	Control	Nominal ohm	Reading	Range	Notes
D1	shorted leads	0			
D1	open leads	OL			
D1	resistor 100 ohm 1 %	100			
D1	resistor 10 kohm 1 %	10000			
D1	resistor 1 Mohm 1 %	1000000			
D1	un-lased film, probes 10 mm apart	OL			
D2	shorted leads	0			
D2	open leads	OL			
D2	resistor 100 ohm 1 %	100			
D2	resistor 10 kohm 1 %	10000			
D2	resistor 1 Mohm 1 %	1000000			
D2	un-lased film, probes 10 mm apart	OL			

Laser-cut holes

Try a 2.00 mm dowel pin (or a 2.00 mm drill shank) in each hole.

Sample	Cut through Y/N	Dowel: slides / tight / no	Notes
E2-B-D1-C1-H1			
E2-B-D1-C1-H2			
E2-B-D1-C1-H3			
E2-B-D1-C2-H1			
E2-B-D1-C2-H2			
E2-B-D1-C2-H3			
E2-B-D2-C1-H1			
E2-B-D2-C1-H2			
E2-B-D2-C1-H3			